

Experiment 6

Aim: To create a stored procedure transfer_employee(emp_id, new_dept_id) with validation and error handling, and to implement database triggers for salary validation and audit logging.

Source Code:

```
-- =====
-- 1. CREATE DATABASE TABLES
-- =====

CREATE TABLE department (
    department_id INT PRIMARY KEY,
    department_name VARCHAR(50) NOT NULL
);

CREATE TABLE employee (
    employee_id INT PRIMARY KEY,
    employee_name VARCHAR(100) NOT NULL,
    department_id INT,
    salary DECIMAL(10,2),
    FOREIGN KEY (department_id) REFERENCES department(department_id)
);

-- Sample departments
INSERT INTO department VALUES
(1, 'IT'),
(2, 'HR'),
(3, 'Finance'),
(4, 'Marketing');

-- Sample employees
INSERT INTO employee VALUES
(101, 'Rahul', 1, 50000),
(102, 'Aman', 2, 45000),
(103, 'Neha', 1, 60000),
(104, 'Priya', 3, 55000),
(105, 'Rohan', 4, 40000);

-- =====
-- 2. CREATE AUDIT TABLE
```

```

-- =====
CREATE TABLE employee_audit (
    audit_id INT AUTO_INCREMENT PRIMARY KEY,
    employee_id INT,
    old_department_id INT,
    new_department_id INT,
    old_salary DECIMAL(10,2),
    new_salary DECIMAL(10,2),
    action_type VARCHAR(50),
    action_time TIMESTAMP DEFAULT CURRENT_TIMESTAMP
);
-- =====

-- 3. STORED PROCEDURE
-- transfer_employee(emp_id, new_dept_id)
-- =====

```

DELIMITER //

```

CREATE PROCEDURE transfer_employee(
    IN p_emp_id INT,
    IN p_new_dept_id INT
)
BEGIN
    DECLARE v_old_dept_id INT;
    DECLARE v_emp_count INT DEFAULT 0;
    DECLARE v_dept_count INT DEFAULT 0;
    -- Error handling
    DECLARE EXIT HANDLER FOR SQLEXCEPTION
    BEGIN
        ROLLBACK;
        SELECT 'Error: Transfer failed.' AS message;
    END;
    START TRANSACTION;
    -- Check employee exists

```

```

SELECT COUNT(*)
INTO v_emp_count
FROM employee
WHERE employee_id = p_emp_id;
IF v_emp_count = 0 THEN
    SIGNAL SQLSTATE '45000'
    SET MESSAGE_TEXT = 'Error: Employee does not exist.';
END IF;
-- Check department exists
SELECT COUNT(*)
INTO v_dept_count
FROM department
WHERE department_id = p_new_dept_id;
IF v_dept_count = 0 THEN
    SIGNAL SQLSTATE '45000'
    SET MESSAGE_TEXT = 'Error: New department does not exist.';
END IF;
-- Get current department
SELECT department_id
INTO v_old_dept_id
FROM employee
WHERE employee_id = p_emp_id;
-- Check if already in department
IF v_old_dept_id = p_new_dept_id THEN
    SIGNAL SQLSTATE '45000'
    SET MESSAGE_TEXT = 'Error: Employee is already in this department.';
END IF;
-- Transfer employee
UPDATE employee
SET department_id = p_new_dept_id
WHERE employee_id = p_emp_id;
COMMIT;
SELECT 'Employee transferred successfully.' AS message;

```

```

END //
DELIMITER ;

-- =====
-- 4. TRIGGER FOR SALARY VALIDATION
-- =====

DELIMITER //
CREATE TRIGGER before_employee_insert
BEFORE INSERT ON employee
FOR EACH ROW
BEGIN
    IF NEW.salary < 0 THEN
        SIGNAL SQLSTATE '45000'
        SET MESSAGE_TEXT = 'Error: Salary cannot be negative.';
    END IF;
END //
DELIMITER ;

-- =====
-- 5. TRIGGER FOR SALARY UPDATE VALIDATION
-- =====

DELIMITER //
CREATE TRIGGER before_employee_salary_update
BEFORE UPDATE ON employee
FOR EACH ROW
BEGIN
    IF NEW.salary < 0 THEN
        SIGNAL SQLSTATE '45000'
        SET MESSAGE_TEXT = 'Error: Salary cannot be negative.';
    END IF;
END //
DELIMITER ;

-- =====
-- 6. AUDIT TRIGGER
-- Logs department transfers
-- =====

```

```

DELIMITER //

CREATE TRIGGER after_employee_update
AFTER UPDATE ON employee
FOR EACH ROW
BEGIN
    IF OLD.department_id <> NEW.department_id THEN
        INSERT INTO employee_audit
        (
            employee_id,
            old_department_id,
            new_department_id,
            old_salary,
            new_salary,
            action_type
        )
        VALUES
        (
            OLD.employee_id,
            OLD.department_id,
            NEW.department_id,
            OLD.salary,
            NEW.salary,
            'DEPARTMENT TRANSFER'
        );
    END IF;
END //

DELIMITER ;

-- =====
-- 7. TESTING
-- =====

-- Test 1: Valid transfer
CALL transfer_employee(101, 3);

-- Check employee
SELECT * FROM employee WHERE employee_id = 101;

```

```

-- Check audit
SELECT * FROM employee_audit;

-- Test 2: Invalid employee
CALL transfer_employee(999, 2);

-- Test 3: Invalid department
CALL transfer_employee(102, 999);

-- Test 4: Employee already belongs to department
CALL transfer_employee(103, 1);

-- Test 5: Negative salary during INSERT
INSERT INTO employee
VALUES (106, 'Test Employee', 1, -5000);

-- Test 6: Negative salary during UPDATE
UPDATE employee
SET salary = -1000
WHERE employee_id = 102;

```

```

-- =====
-- 8. FINAL DATA
-- =====

```

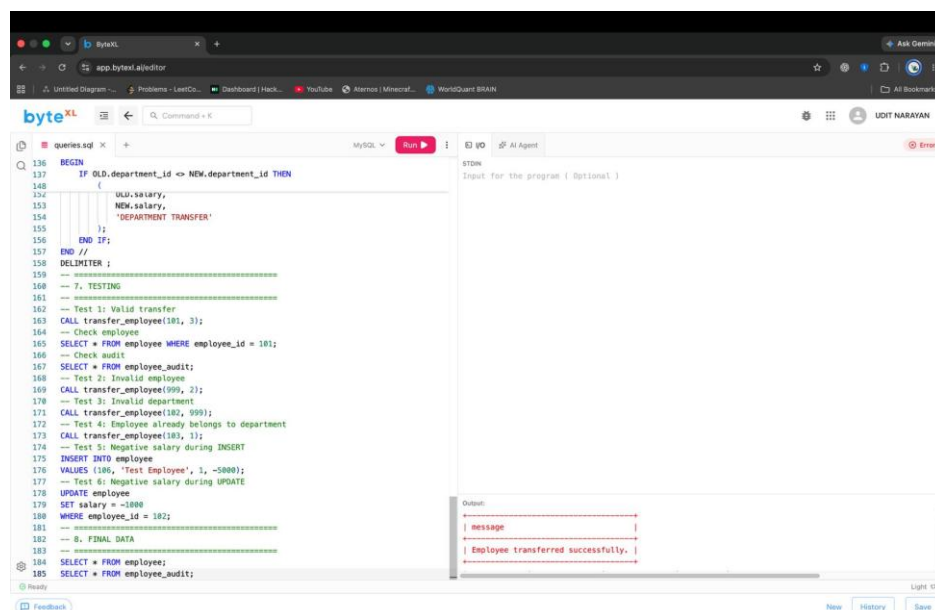
```

SELECT * FROM employee;
SELECT * FROM employee_audit;

```

OUTPUT:

1: SUCCESSFUL EMPLOYEE TRANSFER:



The screenshot shows a MySQL IDE interface with a SQL script in the editor and its output in the right-hand pane. The script includes a stored procedure for transferring employees, followed by several test cases. The output pane shows the results of the tests, indicating that the employee transfer was successful.

```

136 BEGIN
137 IF OLD.department_id <> NEW.department_id THEN
138 (
139     OLD.salary,
140     NEW.salary,
141     'DEPARTMENT TRANSFER'
142 );
143 END IF;
144 END //
145 DELIMITER ;
146 -- =====
147 -- 7. TESTING
148 -- =====
149 -- Test 1: Valid transfer
150 CALL transfer_employee(101, 3);
151 -- Check employee
152 SELECT * FROM employee WHERE employee_id = 101;
153 -- Check audit
154 SELECT * FROM employee_audit;
155 -- Test 2: Invalid employee
156 CALL transfer_employee(999, 2);
157 -- Test 3: Invalid department
158 CALL transfer_employee(102, 999);
159 -- Test 4: Employee already belongs to department
160 CALL transfer_employee(103, 1);
161 -- Test 5: Negative salary during INSERT
162 INSERT INTO employee
163 VALUES (106, 'Test Employee', 1, -5000);
164 -- Test 6: Negative salary during UPDATE
165 UPDATE employee
166 SET salary = -1000
167 WHERE employee_id = 102;
168 -- 8. FINAL DATA
169 -- =====
170 SELECT * FROM employee;
171 SELECT * FROM employee_audit;

```

Output:

```

| message |
| Employee transferred successfully. |

```

2: INVALID EMPLOYEE / DEPARTMENT VALIDATION:

The screenshot shows the byteXL MySQL IDE interface. The SQL editor on the left contains a PL/SQL block with various tests. The output window on the right displays the results of the execution, including a message 'Employee transferred successfully.' and a table showing the audit details of the transfer.

```
136 BEGIN
137   IF OLD.department_id <> NEW.department_id THEN
138     (
139       OLD.salary,
140       NEW.salary,
141       'DEPARTMENT TRANSFER'
142     )
143   END IF;
144 END //
145 DELIMITER ;
146 -- 7. TESTING
147 -- =====
148 -- Test 1: Valid transfer
149 CALL transfer_employee(181, 3);
150 -- Check employee
151 SELECT * FROM employee WHERE employee_id = 181;
152 -- Check audit
153 SELECT * FROM employee_audit;
154 -- Test 2: Invalid employee
155 CALL transfer_employee(999, 2);
156 -- Test 3: Invalid department
157 CALL transfer_employee(182, 999);
158 -- Test 4: Employee already belongs to department
159 CALL transfer_employee(181, 1);
160 -- Test 5: Negative salary during INSERT
161 INSERT INTO employee
162 VALUES (106, 'Test Employee', 1, -50000);
163 -- Test 6: Negative salary during UPDATE
164 UPDATE employee
165 SET salary = -1000
166 WHERE employee_id = 182;
167 -- 8. FINAL DATA
168 -- =====
169 SELECT * FROM employee;
170 SELECT * FROM employee_audit;
```

Output:

```
message
Employee transferred successfully.

employee_id | employee_name | department_id | salary
-----
181 | Rahul | 3 | 50000.00

audit_id | employee_id | old_department_id | new_department_id | old_salary | new_salary | action_type | action_time
-----
1 | 181 | 1 | 3 | 50000.00 | 50000.00 | DEPARTMENT TRANSFER | 2026-10-06 18:22:15
```

3: SALARY VALIDATION TRIGGER:

The screenshot shows the same byteXL MySQL IDE interface as before, but the output window now displays 'Error: Transfer failed.' for the tests involving invalid employee IDs, departments, and negative salaries, indicating that the salary validation trigger is active and preventing these operations.

```
message
Error: Transfer failed.
Error: Transfer failed.
Error: Transfer failed.
Error: Transfer failed.
```

4: SALARY AUDIT LOG:

